

PIOTR MOCZYBRODA

+48 733 335 848 · psmoczybroda@gmail.com · moczybroda.com

Aerospace Engineer (TU Delft) with a BSc thesis on a tube-launched UAV developed in collaboration with the Dutch National Aerospace Laboratory (NLR) – project recognised by the Polish Ministry of Defence. Professional experience as a structural engineer, delivering FEM, fatigue, and dynamic analyses across offshore wind structures and jack-up vessel modifications.

EXPERIENCE

TWD

Rotterdam, Netherlands
Aug 2025 – Aug 2026

Structural Engineer

Delivered structural, fatigue, and dynamic analyses across offshore wind and marine projects, applying FEM and Python-based calculation workflows.

- Completed spudcan modification for a jack-up vessel from concept through detailed phase within 12 weeks with FEM analysis
- Performed signal analysis on accelerometer data to characterise the dynamic behaviour of a structure with failing components
- Delivered a wind turbine blade seafastening structure design within 70% of standard project duration due to client-critical urgency
- Conducted qualitative fatigue studies of offshore seafastening structures

Lambach Aircraft Student Team

Delft, Netherlands
Sep 2023 – May 2024

Aerodynamics Engineer

Evaluated and selected the optimal airfoil from three candidates for a competition aircraft through systematic CFD analysis in ANSYS Fluent, assessing performance across multiple configurations. Results used in manufacturing decisions for the British Model Flying Association entry as part of a TU Delft student team.

Elab Education

Remote
Oct 2023 – Jun 2025

Tutor

Prepared over 20 students for university entrance exams in mathematics, physics, and engineering

EDUCATION

Delft University of Technology

Netherlands
Sep 2022 – Jun 2025

BSc Aerospace Engineering

Specialised in aircraft design with a focus on UAVs for defence applications. Core modules in aerodynamics, structural design, and jet engines

Technical University of Valencia

Spain
Sep 2024 – Jan 2025

Minor Exchange Programme

International exchange covering production engineering, project management, and project budget management.

PROJECTS

BSc Thesis TU Delft x NLR Apr 2025 – Jun 2025	Tube-launched UAV for Surveillance Missions Co-developed a multi-physics simulation model for a tube-launched UAV wing unfolding mechanism, integrating aerodynamic, structural (FEM), and stability analyses across the full deployment phase Delivered in collaboration with the Dutch National Aerospace Laboratory (NLR)
TU Delft Feb 2025 – Apr 2025	Simulation, Verification & Validation Developed and validated a numerical simulation model for Cessna aircraft performance; applied V&V methodology and hypothesis testing to quantify model accuracy against real-world flight data
TU Delft Sep 2023 – Jan 2024	Aircraft Wing Design for Small Passenger Aircraft Executed full wing design cycle: initial sizing, airfoil selection, control surface design, and primary component integration. Delivered wing structure according to CS-23/25 requirements
TU Delft Nov 2023 – Dec 2023	Wind Tunnel Experiment Conducted aerodynamic analysis of 2D and 3D wing configurations using wind tunnel testing and computational simulation. Identified and quantified discrepancies between CFD predictions and experimental lift, drag, and stall data

TECHNICAL SKILLS

Simulation & FEA	RFEM, ANSYS Fluent
CAD	SolidWorks, CATIA
Programming	Python, MATLAB
Standards	ISO, DNV, EASA
Tools & Office	MS Office, LaTeX, Jupyter

LANGUAGES

Polish (Native) · **English** (Full Professional) · **Spanish** (Intermediate) · **German** (Elementary)